Sohyun **Lee**

lshig96@postech.ac.kr | https://sohyun-l.github.io

EDUCATION

POSTECH | *Integrated M.S. · Ph.D.* **Sep. 2020 – Feb. 2026**
- Graduate School of Artificial Intelligence
- Supervised by Prof. Suha Kwak in the Computer Vision Lab.

POSTECH | *B.S.* **March 2015 – Aug. 2020**
- Mechanical Engineering

EXPERIENCE

Postdoctoral Researcher | *Computer Vision Lab, POSTECH* **Mar. 2026 – Present**
- PI: Prof. Suha Kwak

Visiting Researcher (Remote) | *MBZUAI* **Mar. 2026 – June 2026**
- Host: Prof. Ivan Laptev

Research Collaboration | *ETH Zürich* **Aug. 2025 – Jan. 2026**
- Working with Prof. Konrad Schindler and Dr. Christos Sakaridis

Visiting Researcher | *ETH Zürich* **May 2025 – Aug. 2025**
- Host: Prof. Konrad Schindler and Dr. Christos Sakaridis

Research Collaboration | *Google Zürich* **Oct. 2024 – Jan. 2026**
- Working with Dr. Lukas Hoyer

Visiting Researcher | *Tübingen AI Center, University of Tübingen* **Mar. 2024 – May 2024**
- Host: Prof. Seong Joon Oh

Undergraduate Intern | *Innovative Medical Solution Lab, POSTECH* **June 2019 – Sep. 2019**
- Researched on mental stress detection.

Undergraduate Intern | *Industrial AI Lab, POSTECH* **June 2018 – June 2019**
- Researched on lesion detection in capsule endoscopy.

Engineering Intern | *Doosan Heavy Industries* **June 2017 – July 2017**
- Designed a gas turbine compressor.

PUBLICATIONS

International Conference

[1] **Sohyun Lee**, Yeho Gwon, Lukas Hoyer, Konrad Schindler, Christos Sakaridis, and Suha Kwak
Robust Promptable Video Object Segmentation
IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2026

[2] **Sohyun Lee**, Yeho Gwon, Lukas Hoyer, and Suha Kwak
GaRA-SAM: Robustifying Segment Anything Model with Gated-Rank Adaptation
Conference on Neural Information Processing Systems (**NeurIPS**), 2025

[3] **Sohyun Lee**, Namyup Kim, Sungyeon Kim, and Suha Kwak
FREST: Feature RESToration for Semantic Segmentation under Multiple Adverse Conditions
European Conference on Computer Vision (**ECCV**), 2024

[4] Sehyun Hwang, **Sohyun Lee**, Hoyoung Kim, Minhyeon Oh, Jungseul Ok, and Suha Kwak
Active Learning for Semantic Segmentation with Multi-class Label Query
Conference on Neural Information Processing Systems (**NeurIPS**), 2023

[5] **Sohyun Lee***, Jaesung Rim*, Boseung Jeong, Geonu Kim, ByungJu Woo, Haechan Lee, Sunghyun Cho, and Suha Kwak (*equal contribution)
Human Pose Estimation in Extremely Low-light Conditions
IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2023

[6] Sehyun Hwang, **Sohyun Lee**, Sungyeon Kim, Jungseul Ok, and Suha Kwak
Combating Label Distribution Shift for Active Domain Adaptation
European Conference on Computer Vision (**ECCV**), 2022

[7] **Sohyun Lee**, Taeyoung Son, and Suha Kwak
FIFO: Learning Fog-invariant Features for Foggy Scene Segmentation
IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2022
(**Best Paper Finalist, Oral Presentation**)

[8] Juwon Kang, **Sohyun Lee**, Namyup Kim, and Suha Kwak
Style Neophile: Constantly Seeking Novel Styles for Domain Generalization
IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2022

International Journal

[1] **Sohyun Lee**, Nayeong Kim, Juwon Kang, Seong Joon Oh, and Suha Kwak
TestDG: Test-time Domain Generalization for Continual Test-time Adaptation
Transactions on Machine Learning Research (**TMLR**), 2026

[2] Seungwoo Yoon*, Dohyun Kang*, Eunsue Choi, **Sohyun Lee**, Hyeonsu Heo, Minho Choi, Dongha Shin, Seoyeon Kim, Suha Kwak, Arka Marjumdar, Junsuk Rho†, Seung-Hwan Baek†
De-occluding Broadband Metalens
Nature Communications, 2026

Under Review

[1] **Sohyun Lee***, Yoonjae Baek*, Jaesang Won, Jinnyeong Kim, Seung-hwan Baek, Ivan Laptev, Suha Kwak
Robustness to Robot Hardware Imperfections: A Benchmark and a Self-Compensating VLA
Collaboration with MBZUAI
Under review, 2026

ACADEMIC SERVICES

- **Organizer**: Women in Computer Vision Workshop (WiCV) at ACCV 2024.
- **Journal Reviewer**: TPAMI, IJCV
- **Conference Reviewer**: CVPR, CoRL, ICLR, ICML, NeurIPS, ICCV, ECCV, WACV, AAAI, ACCV

HONORS & AWARDS

- KCCV Doctoral Colloquium, 2026
- **CVPR Doctoral Consortium**, 2026
  - Awarded to selected Ph.D. students in computer vision worldwide
- **Best Ph.D Dissertation Award**, College of IT at POSTECH, 2026
  - Awarded to only one Ph.D graduate across 5 departments (CSE, GSAI, EE, CITE, SEMI)
- Excellence Prize at BK21 Best Paper Award, POSTECH GSAI, 2024
- POSTECHIAN fellowship awards, POSTECH, 2023
- Excellence Award at 3rd POSTECH Research Performance Contest, POSTECH, 2023
- Grand Prize at BK21 Best Paper Award, POSTECH GSAI, 2023
- **CVPR Best Paper Finalist**, 2022
  - Awarded to Top 0.4% (33 of 8161 papers)
- **Qualcomm Innovation Fellowship Winner (3 times)** , Qualcomm Korea Corp., 2022
- Gold Prize at IPIU Best Paper Award, 2022
- POSTECH Creative Self-Research Scholarship, 2020

TALK

- Oral presentation, ICCV Workshop on Building Foundation Models You Can Trust, 2025
- LG AI Research Seminar, 2025
- Modulabs Seminar, 2023
- Paper invited talk, Vision for all Seasons Workshop in CVPR, New Orleans, 2022

PATENTS

- U.S. Patent 12,394,084 B2, “Method and Apparatus for Learning Human Pose Estimation in Low-Light Conditions,” granted.
- U.S. Patent 12,579,804 B2, “Method and Device for Learning Fog-Invariant Feature,” granted.

TEACHING ASSISTANT

- Computer Vision, POSTECH
- Reinforcement Learning, POSTECH
- Discrete Mathematics, POSTECH

PRESS

Best Ph.D Dissertation Award, POSTECH

- Feb. 2026, 포스텍, 역대 최대 894명 배출, 뉴데일리
- Feb. 2026, 포스텍 학위수여식, 뉴시스
- Feb. 2026, 포스텍 13일 학위수여식... 894명 학위 취득, 중도일보
- Feb. 2026, POSTECH, 역대 최대 894명 졸업, 대경일보

Foggy Scene Segmentation (CVPR 2022 Best Paper Finalist)

- April 2022, 짙은 안개 껴도 사람·사물 뚜렷이 식별하는 AI 개발, 동아사이언스
- April 2022, 짙은 안개 속에서도 외부환경 정확히 인식하는 AI 개발, 매일경제
- April 2022, 한치 앞도 안 보이는 안개 속에서도 문제없는 자율주행차 나온다, 서울신문
- April 2022, 자율주행車 상용화 앞당긴다...포스텍 연구진, 안개에도 정확한 영상인식 AI기술 개발, 영남일보
- April 2022, 포스텍 곽수하 교수팀 안개 낀 날에도 정확히 동작하는 영상인식 AI 기술 개발, 뉴스1
- April 2022, 안개 낀 날씨에도 정확히 작동 영상인식 AI 개발, YTN사이언스

REFERENCE

- Prof. Suha Kwak, Associate Professor, POSTECH, suha.kwak@postech.ac.kr
- Dr. Christos Sakaridis, Lecturer/Principal Investigator, ETH Zürich, csakarid@ethz.ch
- Dr. Lukas Hoyer, Senior Research Scientist, Google Zürich, lukashoyer@google.com
- Prof. Konrad Schindler, Full Professor, ETH Zürich, schindler@ethz.ch